ME 230 - Lecture Notes: The First Law of Thermodynamics

Chapter 2 | Week 3

1. Systems and State Functions

A thermodynamic system is the region of space we analyze; everything else is the surroundings. A closed system exchanges energy but not mass with its surroundings; an open system (control volume) exchanges both.

State functions depend only on the current equilibrium state of the system, not on how it got there. Internal energy U, enthalpy H, entropy S, pressure P, volume V, and temperature T are state functions. Heat Q and work W are path functions: their values depend on the process path between two states.

2. The First Law of Thermodynamics

For a closed system, energy is conserved: Delta U = Q - W, where Q is the heat added TO the system and W is the work done BY the system on its surroundings. (Some texts use W as work done ON the system and write Delta U = Q + W; always check the sign convention.)

For a cycle the system returns to its initial state, so Delta U = 0 and the net heat added equals the net work done: Q_net = W_net.

3. Work in Ideal-Gas Processes

Boundary work is W = integral of P dV. For an ideal gas PV = nRT (R = 8.314 J/(mol K)).

Isobaric (constant P): W = P (V2 - V1).

Isochoric (constant V): W = 0, so Delta U = Q.

Isothermal (constant T): Delta U = 0 for an ideal gas, and W = nRT ln(V2/V1) = Q.

Adiabatic (Q = 0): P V^gamma = constant with gamma = cp/cv, and W = (P1 V1 - P2 V2)/(gamma - 1) = -Delta U.

Process	Held constant	Work W	Heat Q
Isobaric	P	P(V2 - V1)	n cp Delta T
Isochoric	V	0	n cv Delta T
Isothermal	T	nRT ln(V2/V1)	W
Adiabatic	Q = 0	(P1V1 - P2V2)/(gamma-1)	0

4. Enthalpy and Specific Heats

Enthalpy is defined as H = U + PV. At constant pressure, the heat added equals the change in enthalpy: Q_p = Delta H.

For an ideal gas: Delta U = n cv Delta T and Delta H = n cp Delta T, with cp - cv = R. For a

monatomic ideal gas cv = (3/2)R and cp = (5/2)R, so gamma = 5/3; for a diatomic gas near room temperature cv = (5/2)R and gamma = 1.4.

5. Heat Engines and Efficiency

A heat engine takes heat Q_H from a hot reservoir, does work W, and rejects heat Q_C to a cold reservoir. Its thermal efficiency is eta = W / Q_H = 1 - Q_C / Q_H.

No engine operating between reservoirs at T_H and T_C (in kelvin) can beat the Carnot efficiency eta_Carnot = 1 - T_C / T_H. A real automotive engine is typically 25-35% efficient.

The Otto cycle (idealized gasoline engine) consists of: (1) adiabatic compression, (2) constant-volume heat addition, (3) adiabatic expansion (power stroke), and (4) constant-volume heat rejection. Its efficiency is eta_Otto = 1 - 1 / r^(gamma - 1), where r = V_max / V_min is the compression ratio.

6. Worked Example

Worked example. One mole of an ideal diatomic gas expands isothermally at 300 K from 10 L to 20 L.

Step 1: Isothermal, so Delta U = 0.

Step 2: W = nRT ln(V2/V1) = (1)(8.314)(300) ln(2) = 1729 J.

Step 3: First law: Q = Delta U + W = 1729 J of heat is absorbed.

Common mistake: using V2 - V1 instead of the ratio V2/V1, or forgetting that temperature must be in kelvin.